Omar Abuhammoud

+1 (347)-362-2044 ● oabuhammoud@gmail.com ● LinkedIn ● GitHub ● Portfolio

Education

Cornell University, College of Engineering May 2026
Bachelor of Science in Computer Science, Dean's List

Related Coursework: Algorithms, Data Structures, Artificial Intelligence, Machine Learning, Systems Programming, Object-Oriented Programming, Functional Programming, Linear Algebra, Statistics

Experience

Software Engineer @ Bloomberg LP Website | July 2026-Present
- **Engineered** a Python microservice and AI agent toolset that gives Large Language Models (LLMs) secure, **user-scoped access to enterprise data**, enabling search, retrieval and navigation of chats.
- **Designed** an opaque reference system backed by a replicated database, **keeping internal identifiers out of model context** while guaranteeing consistency across distributed systems.
- **Implemented** request-level authorization and identity resolution to enforce per-user access controls and return accurate, readable results to AI clients.

Robotics Lead @ Cornell Cup Robotics Team Website | February 2024-May 2026
- Developed a modular educational robot for nationwide school distribution that **enhances STEM education** with user-programmable controls and interactive features
- Engineered a lab assistant robot with voice recognition, object detection, and sentiment analysis, **enabling semi-autonomous task** completion.
- Spearheaded the development of a bipedal walking robot using **neural networks** for movement learning and enabling users to control the robot's actions through **Virtual Reality (VR) integration**.

Software Engineer Intern @ Bloomberg LP Website | June 2025-August 2025
- **Engineered** a C++ backend service in a microservices architecture to **reconcile state across multiple data sources**, enhancing consistency and reliability of enterprise communication workflows.
- **Integrated** external APIs, internal databases, and event-driven systems (Kafka, service-to-service RPC) to ensure accurate, **real-time synchronization** of user and room data.
- **Architected** automated validation pipelines, monitoring tools, and scheduled recovery workflows to proactively mitigate data or event gaps and **maintain system robustness**.

Full Stack Intern @ Chapter One Website | October 2024-November 2024
- Collaborated on an AI-driven platform that **generates customizable websites** for new businesses, enhancing digital accessibility and streamlining their online presence.
- **Optimized API performance,** implementing redundancy measures to ensure seamless data transfer, reliability, and faster response times.
- **Enhanced website personalization** by **incorporating custom effects** and **e-commerce capabilities**, allowing businesses to showcase products and sell services on their platforms.

Projects

Networked Multiplayer Party Game, Hard to Herd Link | January 2026-Present
- **Directed a 10-person agile team**, overseeing task delegation, comprehensive code reviews, and version control pipelines to **ship production-ready releases to a public user base**.
- Engineered core multiplayer netcode, spatial partitioning for physics, and a robust file parsing logic for dynamic level loading, ensuring a **seamless and synchronized gameplay experience**.
- Developed custom level editors, dynamic UI components, and core gameplay mechanics including **complex entity spawn systems and state-managed item limitations** to balance match pacing.

Skills

- **Back-end Development Tools:** Java, C/C++, Python, C#, OCaml, SQL
- **Front-end Development Tools:** HTML, JavaScript, CSS, React
- **Developer Tools:** Git, GitHub, AWS